

TOPIC 5: RATIONAL AND NON-RATIONAL NUMBERS

1. Rational Numbers

A rational number can be expressed as a fraction p/q where p and q are integers and $q \neq 0$. Examples: 3, -2, $1/2$, 0.75 (which is $3/4$), $0.333\ldots$ ($1/3$). All integers and all terminating or repeating decimals are rational.

2. Non-Rational (Irrational) Numbers

An irrational number cannot be written as a simple fraction p/q . Its decimal goes on forever without repeating. Examples: $\sqrt{2}$, $\sqrt{3}$, π (pi). $\sqrt{4} = 2$ is rational; $\sqrt{2}$ is not.

3. Identifying Them

If you can write it as a fraction of integers, it is rational. If it is a non-repeating, non-terminating decimal (or a root that does not simplify to an integer or fraction), it is irrational.

4. Summary

Rational numbers are fractions of integers; irrational numbers are not. Together with zero and negatives they form the real numbers used in school mathematics.

End of Topic 5